Task 1 Numerical Methods

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PROBLEM 1

1.2.7

The big-O notation $f(n) = O(g(n))$ means that there exists a constant $C > 0$ and a value n_0 such that $f(n) \leq C \cdot g(n)$ for all $n > n_0$. I will consider each of the assertions:

a. $\frac{n+1}{n^2} = O\left(\frac{1}{n}\right)$

We have:

$$\lim_{n\to\infty} \frac{\frac{n+1}{n^2}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{n^2}{n+1} = \lim_{n\to\infty} \frac{n}{1+\frac{1}{n}} = \infty$$

Since the limit is infinite, $\frac{n+1}{n^2}1$ is not $O\left(\frac{1}{n}\right)$. Therefore **not correct**

b. $\frac{n+1}{\sqrt{n}} = O(1)$

For $n > 1$, we can say $\frac{n+1}{\sqrt{n}} \leq \frac{n+n}{\sqrt{n}} = \frac{2n}{\sqrt{n}} = 2\sqrt{n}$. Since $2\sqrt{n}$ is not bounded as $n \to \infty$, this statement is **not correct**.

c. $\frac{1}{\ln n} = O\left(\frac{1}{n}\right)$

We have:

$$\lim_{n\to\infty} \frac{\frac{1}{\ln n}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{n}{\ln n} = \infty$$

Since the limit is infinite, the statement is **not correct**.

d. $\frac{1}{n \ln n} = O\left(\frac{1}{n}\right)$

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We have:

$$\lim_{n\to\infty} \frac{\frac{1}{n \ln n}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{1}{\ln n} = 0$$

Since the limit is 0 , the statement **is correct**.

e. $\frac{e^n}{n^5} = O\left(\frac{1}{n}\right)$

We have:

$$\lim_{n\to\infty} \frac{\frac{e^n}{n^5}}{\frac{1}{n}} = \lim_{n\to\infty} n \cdot \frac{e^n}{n^5} = \lim_{n\to\infty} \frac{e^n}{n^4} = \infty$$

Since the limit is infinite, the statement is **not correct**.

1.2.35

The statement "If $x_n = O(\alpha_n)$, then $x_n = O(\alpha_n)$ " is a tautology, meaning it is trivially true by definition. As for the converse, if taken literally, it is also a tautology and thus true. However, if the converse is interpreted as "If $x_n = o(\alpha_n)$, then $x_n = O(\alpha_n)$," it is also true. This is because $o(\alpha_n)$ implies that x_n grows much more slowly than α_n, which is a subset of the conditions for $O(\alpha_n)$. Therefore, the converse, under this interpretation, is also true.

1.2.36

To prove $x_n = O(\alpha_n)$ and $y_n = O(\alpha_n)$ implies $x_n + y_n = O(\alpha_n)$, I consider the following:

(1) $x_n = O(\alpha_n)$ implies that $\exists C_1, n_0$ such that $|x_n| \leq C_1|\alpha_n|$ for $n \geq n_0$.
(2) $y_n = O(\alpha_n)$ implies that $\exists C_2, n_0$ such that $|y_n| \leq C_2|\alpha_n|$ for $n \geq n_0$.

We aim to find C and n_0 such that $|x_n + y_n| \leq C|\alpha_n|$ for $n \geq n_0$.

Using the triangle inequality, $|x_n + y_n| \leq |x_n| + |y_n| \leq C_1|\alpha_n| + C_2|\alpha_n| = (C_1 + C_2)|\alpha_n|$.

Thus, take $C = C_1 + C_2$ and n_0 as the maximum n_0 from the individual cases. This proves that $x_n + y_n = O(\alpha_n)$.

2.1.39

I can represent repeating decimals as fractions using algebra. Below i have made the calculations for each case:
Case 0.181818 . . .
Let $x = 0.181818\ldots$. I Multiply by 100 and subtract x to get:

$$99x = 18 \quad \Rightarrow \quad x = \frac{18}{99} = \frac{2}{11}$$

Case 2.7027027027 . . .
Let $x = 2.7027027027\ldots$. I Multiply by 1000 and subtract x to get:

$$999x = 2700 \quad \Rightarrow \quad x = \frac{2700}{999} = \frac{100}{37}$$

Case 98.1981981981 . . .

Let $x = 98.1981981981\ldots$. I Multiply by 1000 and subtract x to get:

$$999x = 98100 \quad \Rightarrow \quad x = \frac{98100}{999} = \frac{10900}{111}$$

2.2.8

To address the loss of significance in solving $ax^2 + bx + c = 0$ using the quadratic formula, we modify the formula using a similar technique as shown in Example 1 on page 56.
For the first root, let's consider:

$$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

We multiply and divide by the conjugate of the numerator to get:

$$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \times \frac{-b - \sqrt{b^2 - 4ac}}{-b - \sqrt{b^2 - 4ac}}$$
$$x_1 = \frac{-b^2 + (b^2 - 4ac)}{-2a(b + \sqrt{b^2 - 4ac})}$$
$$x_1 = \frac{-4ac}{-2a(b + \sqrt{b^2 - 4ac})}$$
$$x_1 = \frac{2c}{b + \sqrt{b^2 - 4ac}}$$

This new form avoids the loss of significance when $4ac \ll b^2$.
For the other root, we can calculate it as:

$$x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

And since $x_1 \times x_2 = \frac{c}{a}$,

$$x_2 = \frac{c}{ax_1}$$

So to avoid loss of significance, we can use:

$$x_1 = \frac{2c}{b + \sqrt{b^2 - 4ac}}$$
$$x_2 = \frac{c}{ax_1}$$

PROBLEM 2

The problem asks us to find a value ε such that the approximation $\cos(x) \approx 1$ holds within the given accuracy $\frac{1}{2} \times 10^{-8}$ for $|x| < \varepsilon$. We can use the Taylor series expansion of $\cos(x)$ to analyze the error. The Taylor series of $\cos(x)$ centered at $x = 0$ is:

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

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Truncating this series after the first term, we get $\cos(x) \approx 1$. The error term $E(x)$ can be found as:

$$E(x) = \cos(x) - 1 = -\frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

We are interested in finding ε such that $|E(x)| < \frac{1}{2} \times 10^{-8}$ when $|x| < \varepsilon$. To bound the error, I use the first term:

$$|E(x)| = \left|\frac{-x^2}{2!}\right| = \frac{x^2}{2}$$

I want this to be less than $\frac{1}{2} \times 10^{-8}$:

$$\frac{x^2}{2} < \frac{1}{2} \times 10^{-8}$$

Solving this inequality, gives me:

$$x^2 < 10^{-8}$$

$$|x| < 10^{-4}$$

Thus, $\varepsilon = 10^{-4}$.

For $|x| < 10^{-4}$, the approximation $\cos(x) \approx 1$ should be accurate to within $\frac{1}{2} \times 10^{-8}$. The function $f(x)$ can then be defined as:

$$f(x) = \begin{cases} 1 & ; |x| < 10^{-4} \\ \cos(x) - \sin(x) & ; |x| \geq 10^{-4} \end{cases}$$

PROBLEM 3

117 in Decimal to Binary:

117 ÷ 2 = 58 remainder 1

58 ÷ 2 = 29 remainder 0

29 ÷ 2 = 14 remainder 1

14 ÷ 2 = 7 remainder 0

7 ÷ 2 = 3 remainder 1

3 ÷ 2 = 1 remainder 1

1 ÷ 2 = 0 remainder 1

If we read the remainders from bottom to top, we get 1110101.

149 in Decimal to Binary:

149 ÷ 2 = 74 remainder 1

74 ÷ 2 = 37 remainder 0

37 ÷ 2 = 18 remainder 1

18 ÷ 2 = 9 remainder 0

9 ÷ 2 = 4 remainder 1

4 ÷ 2 = 2 remainder 0

2 ÷ 2 = 1 remainder 0

1 ÷ 2 = 0 remainder 1

Reading the remainders from bottom to top, we get ' 10010101 '.

For the fractional part, '.7' in this case for 149.7 , we can use multiplication by 2:

$0.7 \times 2 = 1.4$

$0.4 \times 2 = 0.8$

$0.8 \times 2 = 1.6$

$0.6 \times 2 = 1.2$

$0.2 \times 2 = 0.4$

We can continue this process to get more precision, but I'll stop here. Concatenating these gives me ' 10110̇ for the fractional part. Therefore 149.7 in binary is approximately 10010101.10110

PROBLEM 4

A) The largest number that can be represented in the Marc-32 system is 3.4×10^4, which is less than 10^{40}. Therefore, 10^{40} is not a Marc-32 number.

B) The number $2^{-1} + 2^{-29}$ translates to approximately 0.50000000186 when converted to decimal form. Since Marc-32 can only accommodate up to 6 decimal places, this number cannot be accurately represented and is thus not a Marc- 32 number.

C) Writing $\frac{1}{3}$ in decimal form reveals that it has an infinite repeating sequence. This exceeds the number of bits that can be allocated in the Marc-32 system. Therefore, $\frac{1}{3}$ is not a Marc32 number.

D) Likewise, converting $\frac{1}{5}$ to its decimal form yields 0.2 , which when translated into binary becomes an infinite repeating sequence $(0.\overline{0011})_2$. This also exceeds the number of bits that can be allocated in the Marc-32 system, making $\frac{1}{5}$ not a Marc-32 number as well.

This answer effectively addresses the problem by scrutinizing each of the given numbers to determine whether they can be accurately represented in the Marc-32 system, taking into account the specific limitations of the system.

PROBLEM 5

In order to convert decimal numbers to a specialized Marc-32 format, I implemented a Python script. The program performs several key steps: it calculates the sign bit based on the number's sign, converts both the integer and fractional parts to binary, normalizes the number, calculates a biased exponent, and finally generates the mantissa. These steps are modularized into specific Python functions.

Using this approach, the following Marc-32 representations were obtained for the given decimal numbers:

For 11.5 , the representation is ' $\mathbf{0 \times 13018000000}$ '

For 1.0625 , the representation is ' $\mathbf{0 \times 12701000000000000000000}$ '

For 2.0009765625 , the representation is ' $\mathbf{0 \times 12800000010000000000000}$ '

For 14.52 , the representation is ' $0 \times 131851EB852$